

A black and white photograph of the TIAGo robot head, which has a square face with two circular eyes and a small sensor array on top. The PAL logo is visible on the front of the head.

PAL

TIAGo Head: an AI Powered Platform for Social Robotics

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Why a tabletop social robot?

- **Social robots** enable natural, face-to-face interaction in education, healthcare, public services, and research.
- **Tabletop robots** are compact, stationary, lower-cost and engaging robots — perfect for:



Education



Healthcare & Elderly Care



Customer Service



HRI Research

- Many are **closed platforms** (limited customization)

Our goal: To create a **flexible, open-source, expressive, AI-enabled** tabletop robot to advance research and real-world deployment.

Where Does TIAGo Head Fit?

Open ROS 2 platform with modular architecture

Runs local or cloud LLMs for natural dialogue

Integrated multimodal expressiveness (face, head, LEDs, gestures)

Designed for both research flexibility and deployable applications



Hardware overview



DIMENSIONS

Height
45 cm

Width
32 cm

Depth
25.5 cm

Weight
5.75 kg

DEGREES OF FREEDOM

Neck
2 (Pan and Tilt)

POWER SUPPLY

Mains
230V/110V AC (no
battery)

HARDWARE

Touchscreen
10" Capacitive touchscreen
LEDs
4 programmable RGB zones

Camera
RealSense RGB-D camera

Microphone
4 -mic microphone array with
hardware audio preprocessing
(reSpeaker)

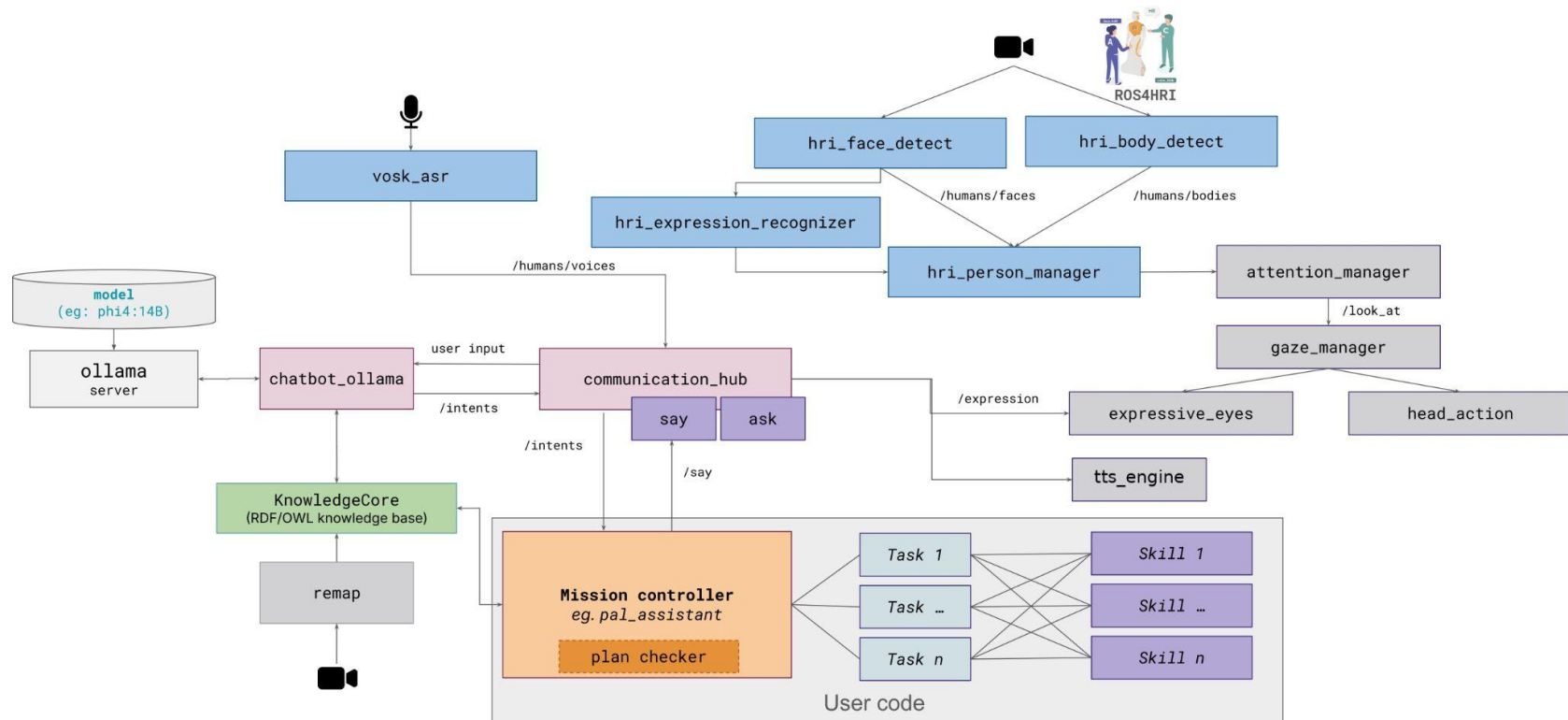
Speakers
2× 4W speakers

Computer
ITX with i7, 16 GB RAM, 500 GB SSD

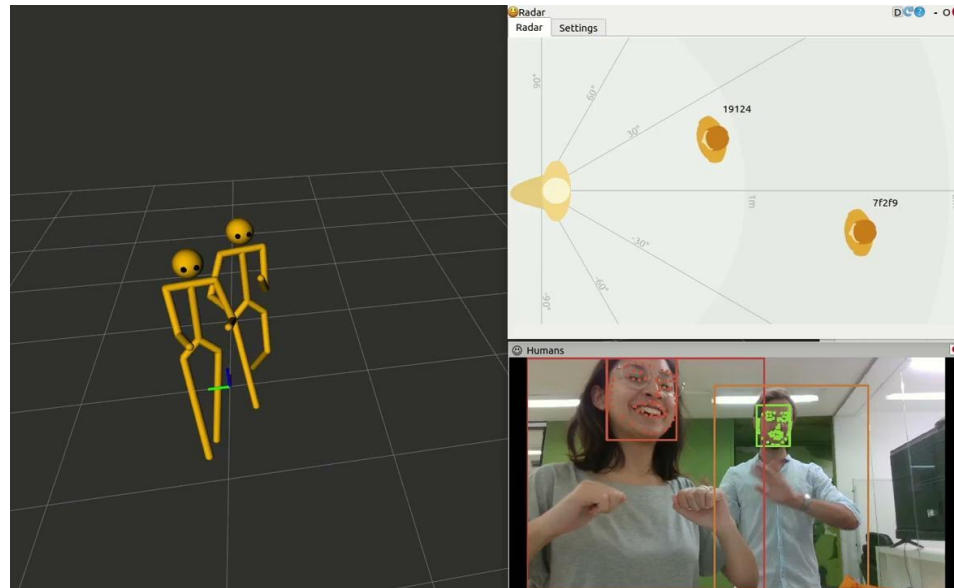
Optional GPU
Nvidia Jetson Orin AGX

Optional tablet (QML)

Software architecture



1 Perception and social sensing



Open-source:
<https://github.com/ros4hri>

`hri_face_detect`: YuNet face
detector + Mediapipe face mesh

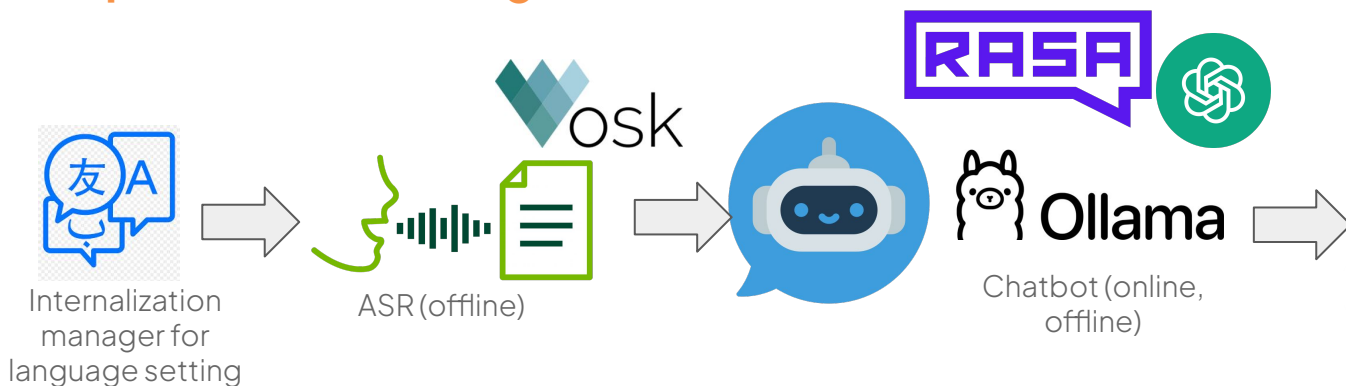
`hri_expression_recognizer`:
ONNX Ferplus model



`hri_person_manager`:
merge face, body for unique ID

`hri_body_detect`: Mediapipe 3D
body pose estimation

2 Speech & LLM Integration



communication_hub
say
ask

Model	Hardware	Gen speed (token/s)	Use case
LLaMA3.2 1B	Intel i7 CPU	24.8	Fast local dialogue
Qwen1.5B	Intel i7 CPU	10.0	Medium local
LLaMA3.2 3B	Intel i7 CPU	3.7	Richer local (slower)
LLaMA3.2 1B	Orin AGX GPU	34.4	Fast + GPU
Phi-4 14B	Orin AGX GPU	9.7	Complex responses

4 Expressiveness & Attention Manager

Communication hub:

Supports markups to sync facial expressions & gestures with speech

Expressions:

Set by name or valence/arousal

Attention manager:

Gaze control for policies
RANDOM, IDLE_SOCIAL,
DISABLED

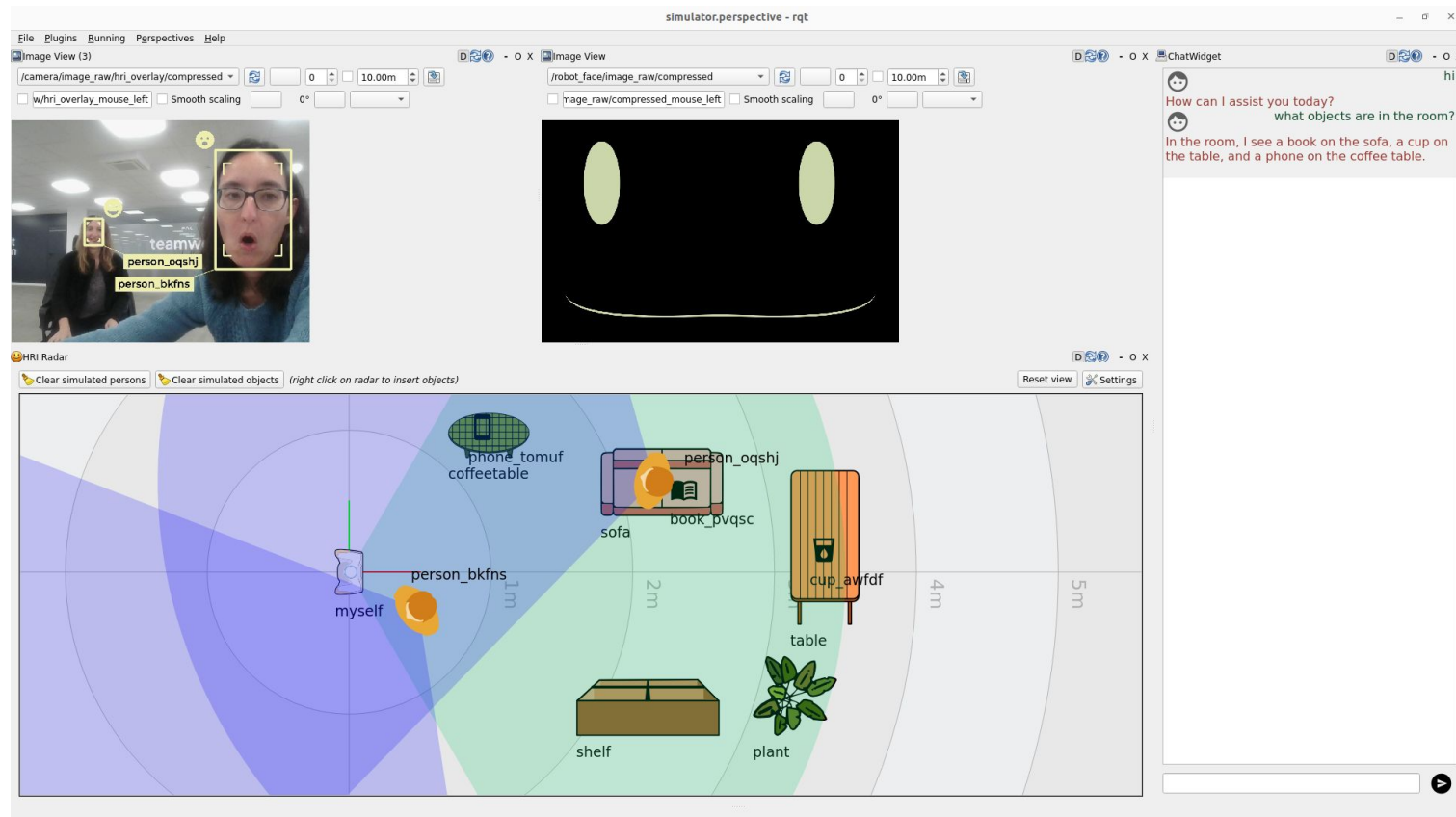


```
<do motion(shake_head)>  
<set expression(confused)> Sorry, I completely forgot your name... was it Sara?  
<set expression(skeptical)>
```

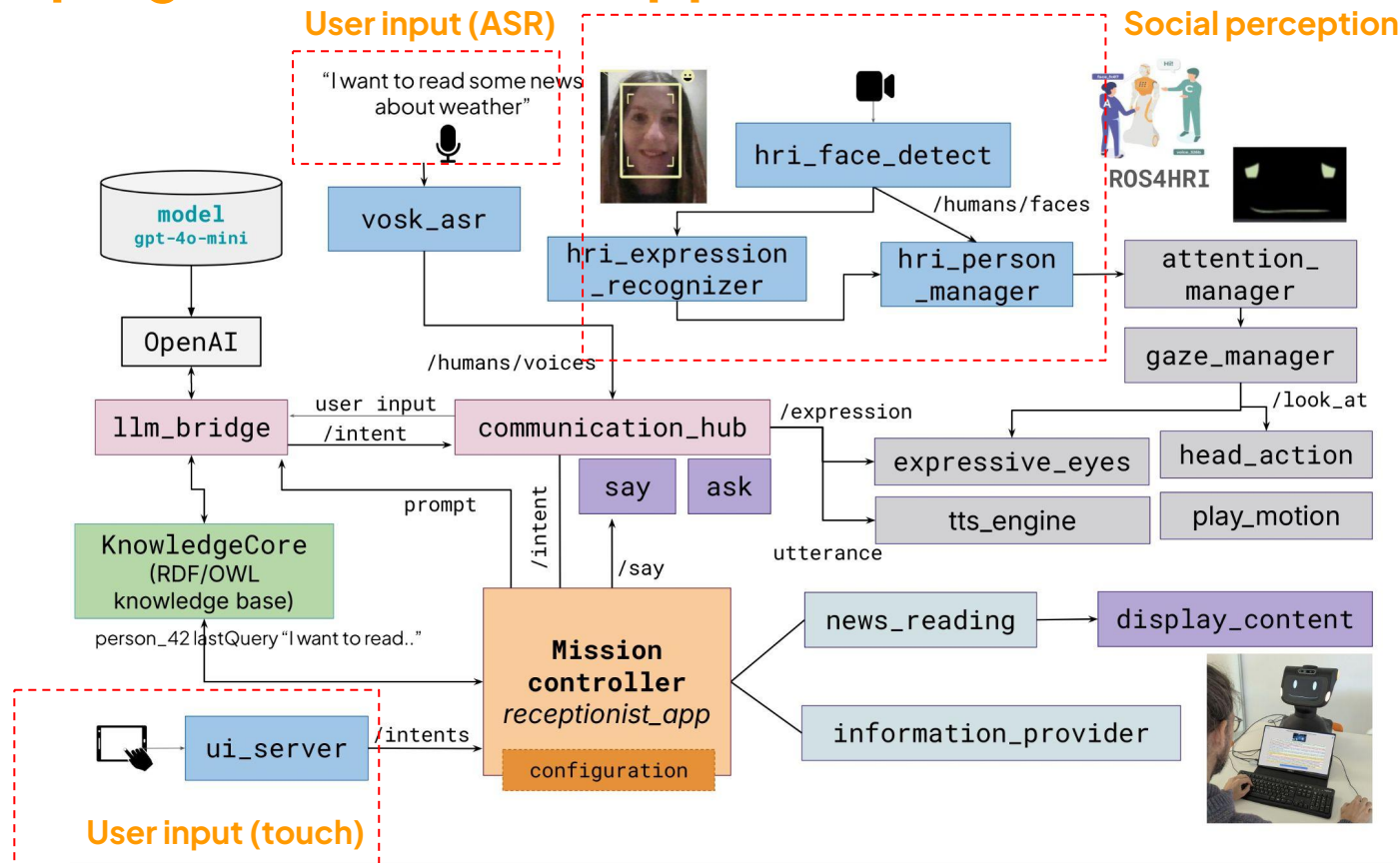


Head action:
perform head motions on 2DoF

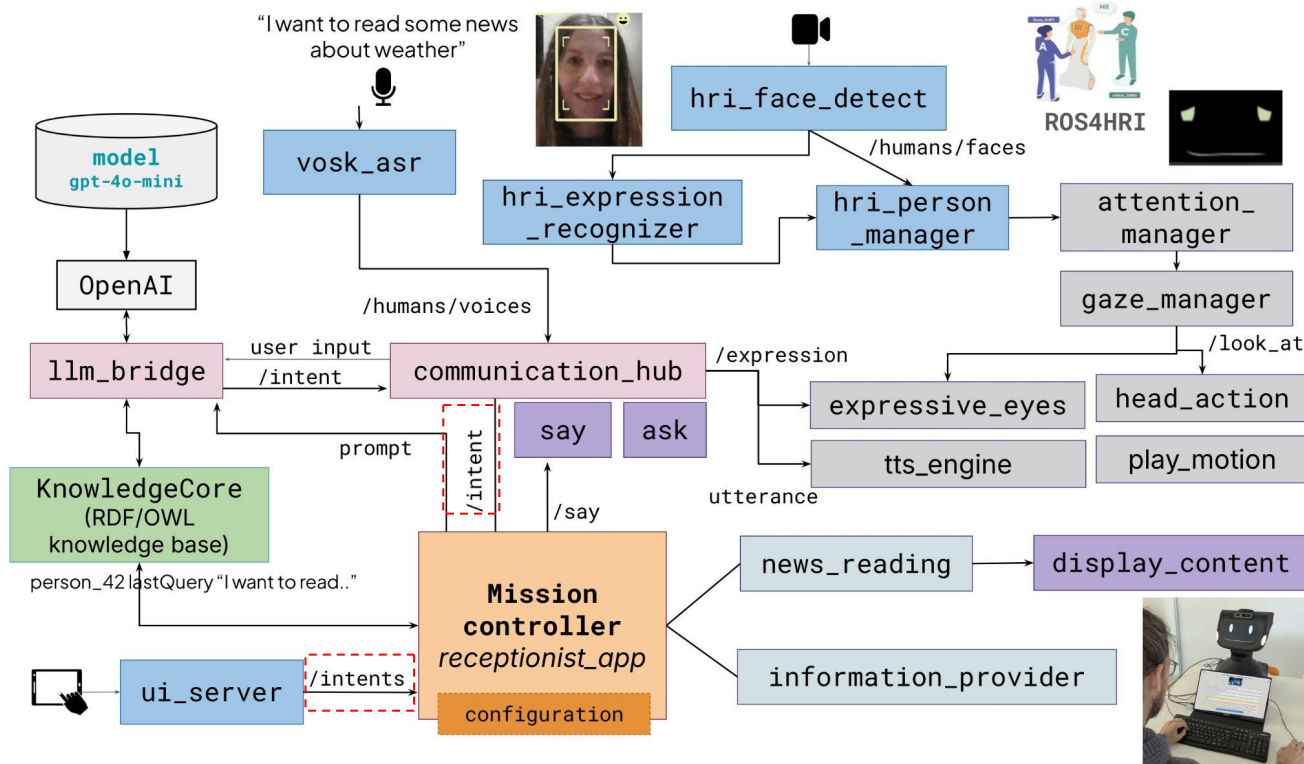
5 Mix-Reality Interaction Simulator



Developing a TIAGo Head Application

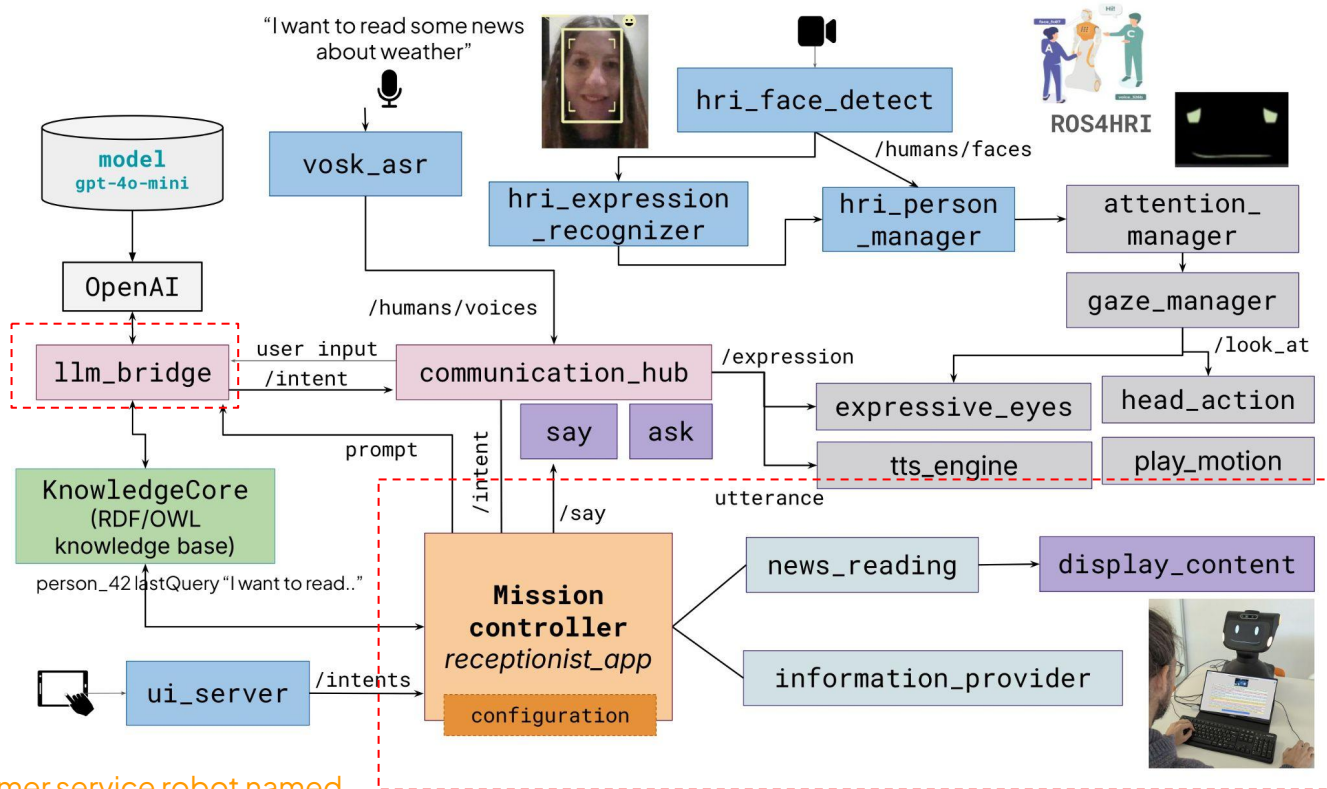


Developing a TIAGo Head Application



User intent detection

Developing a TIAGo Head Application



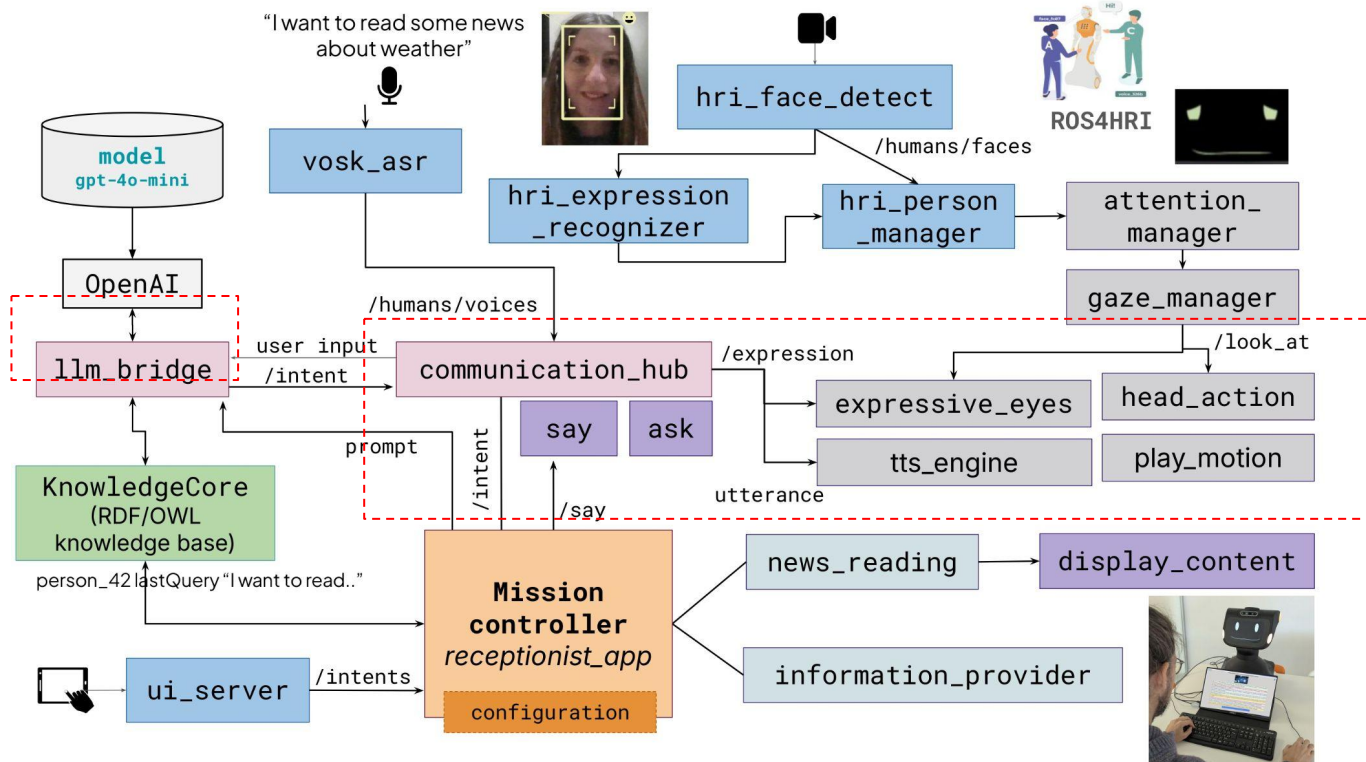
"You are a customer service robot named {{robot name}} for RoboPlane. Your job is

RPK template

2 tasks

1 skill

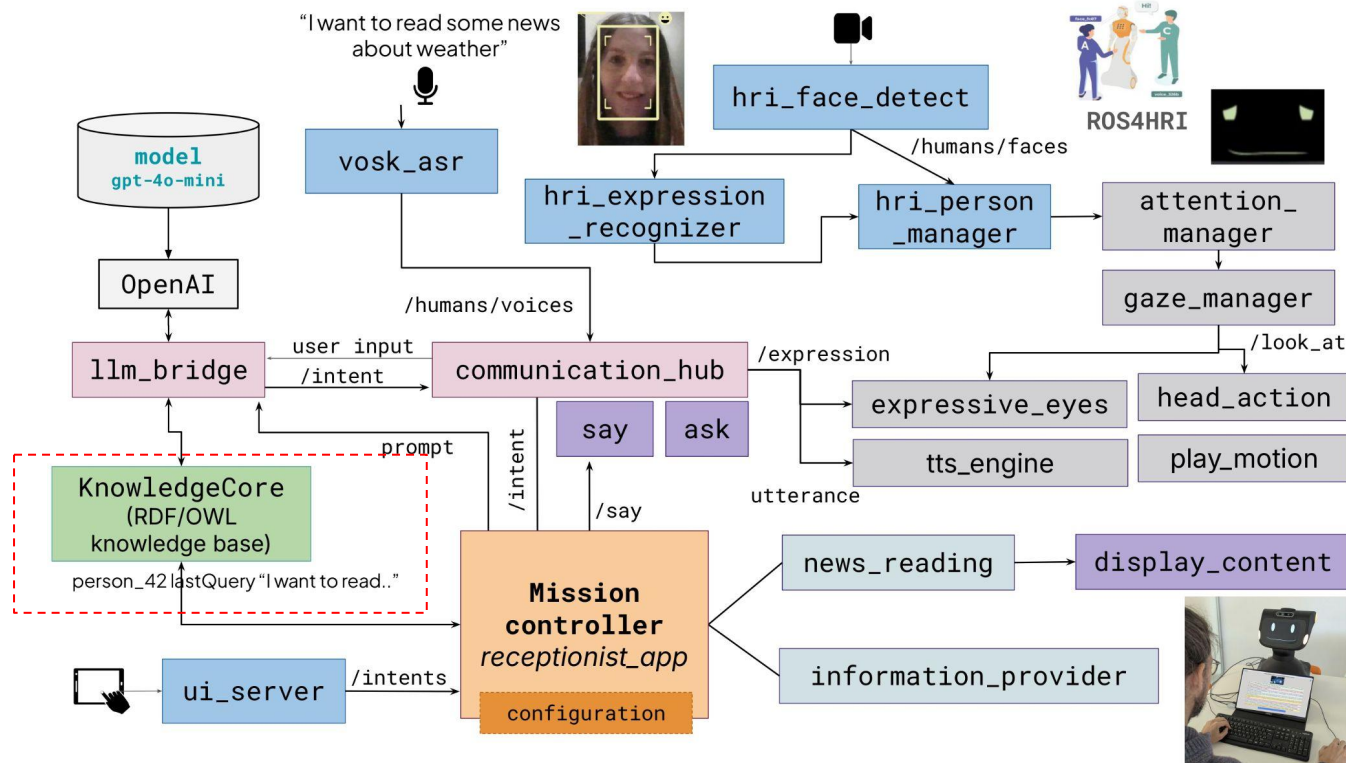
Developing a TIAGo Head Application



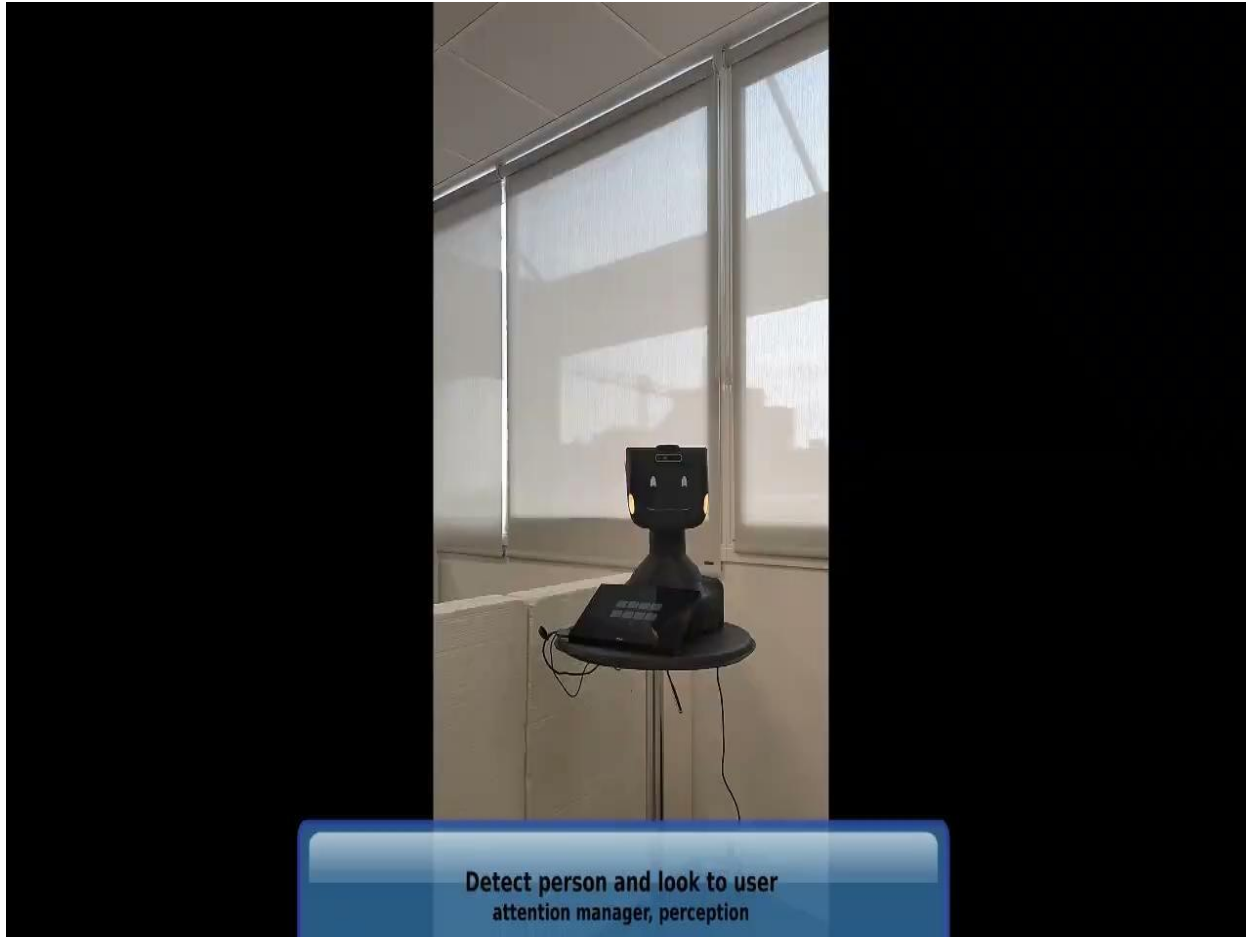
Execute tasks

Speech output: "<set expression(neutral)> Here's the latest travel news: <do motion(head nod)> A major storm in Europe is causing flight delays..."

Developing a TIAGo Head Application



Update knowledge base



Detect person and look to user
attention manager, perception

Conclusions and Future Work

Achievements ✓

- TIAGo Head: Open & Flexible Platform for social HRI research
- ROS 2-based design with facial expressiveness + LLM integration
- Interactive simulator & RPK for rapid application development
- Example scenario: news provider



Next steps and Challenges 🚀

- On-device AI limits → explore knowledge graphs & optimized models
- Privacy & ethics: add real-time filtering & consent management
- Physical deployment & pilot user studies planned
- Re-use open-source software architecture on novel robots

thank you



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